Residual TFT-EWC: Continual Learning for State-of-Health Forecasting and Reuse Screening of Heterogeneous Second-Life EV Batteries

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***Abstract*—Batteries retired from electric vehicles can often be reused as stationary storage, yet a model that judges their health on one group of batteries may fail on the next. Continual learning lets a model learn each new group without forgetting the earlier ones. We propose Residual TFT-EWC, a Temporal Fusion Transformer trained with Elastic Weight Consolidation (EWC) that learns a scaled-down correction to a straight-line extrapolation of the recent fade instead of forecasting the state of health (SOH) from scratch. We forecast the SOH of lithium-ion cells one to six capacity checks ahead on 606 cells from six public datasets, with experiments registered in advance and every claimed difference repeated on a second set of random seeds. Residual TFT-EWC is the only model compared that does well both when groups arrive one after another and on unseen usage conditions. In sequence it matches linear extrapolation, the strongest baseline, is 0.13–0.41 percentage points (pp) more accurate than gradient boosting, and EWC cuts its forgetting by about two thirds. On unseen conditions it is 0.43–0.68 pp more accurate than extrapolation and puts 95.5% of samples in their measured capacity-based health class. Gradient boosting and a jointly trained TFT do better there because they keep every earlier condition's data; denied it, gradient boosting comes within 0.08 pp on unseen conditions but forgets on the stream, where it is 0.24–0.46 pp less accurate, and over 60–64 unseen conditions of two further datasets it is 0.8–1.3 pp less accurate. Forecast intervals reach their stated coverage on a new condition only when calibrated on conditions left out of training. A post hoc variant with replay of 32 stored rows per condition, a three-seed ensemble and calibration on forecasts made before training shows no replicated difference from the joint TFT on those conditions, with 90% coverage of 0.89–0.91, and beats gradient boosting with all data stored on one dataset, mainly where boosting fails, but not on the other.**

***Index Terms*—Continual learning, second-life batteries, electric vehicles, state of health, battery degradation, heterogeneous batteries, catastrophic forgetting, machine learning**

I. Introduction

When an electric-vehicle (EV) battery can no longer meet the demands of driving, it can often still serve as stationary storage or backup power. Reusing it extends its life and delays recycling, but only if its health can be judged reliably.

The state of health (SOH) compares a battery's present capability with a reference state. Machine-learning models have estimated it from voltage, current, temperature, partial-charging and impedance measurements [1]–[4]. Retired batteries, however, differ in chemistry, maker, capacity, past use, temperature and charging, and their records from first use are often incomplete, so a model trained once may fail on the next group. Continual learning updates a model with each new group while trying to keep what it learned before.

This work brings together continual learning, health forecasting and a first screening step for reuse. Rather than estimate today's SOH, we *forecast* it one to six capacity checks ahead from everything measured up to the latest check. On our first dataset, estimating the current SOH from cycling data did worse than simply repeating the last measured value (§III), and a reuse decision depends on how a battery will age, not only on its health today. The forecast is then sorted into capacity-based classes that support, but do not replace, a full reuse assessment.

Fig. 1 summarises the approach, and the paper makes four contributions. First, we build a benchmark for forecasting the SOH of second-life cells from six public datasets (606 cells after quality checks) so that no future information reaches the inputs, and we remove two label flaws: test currents set from the previous capacity, and pulse charge counted as capacity.

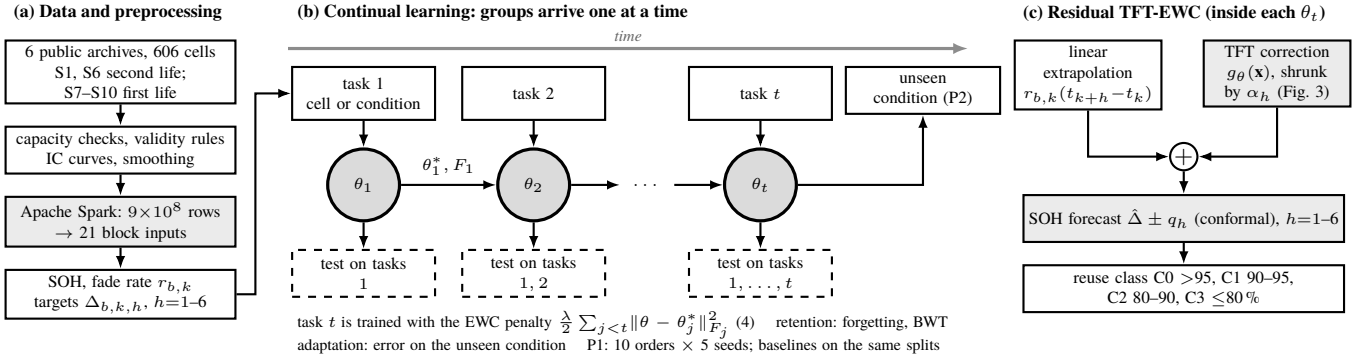


Fig. 1. System architecture. (a) Archives to labels and inputs. (b) Groups arrive as tasks; each state θ_t is trained with EWC and tested on all tasks so far and on unseen conditions. (c) The forecast of each state.

Second, we compare a Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) with fine-tuning, replay, joint training, simple forecasters and the model types of earlier work when groups arrive in sequence, when groups are unseen and after first-life pretraining, reporting both adaptation and forgetting. Third, we propose Residual TFT-EWC, which corrects linear extrapolation instead of replacing it. Among methods that keep no earlier data, it is the only one close to the best in both settings: it matches extrapolation on the stream, where EWC cuts its forgetting by about two thirds, outperforms it on unseen usage conditions, and beats gradient boosting refitted on each new condition on the stream (replicated). Keeping 40 rows per condition, a post hoc replay variant shows no replicated difference from joint training on two many-condition datasets; gradient boosting with all data stored stays better on one. Finally, we give forecast intervals for unseen conditions, which reach the stated coverage only when calibrated on conditions left out of training and within the training range, and a reproducibility check: because significance on these small datasets changed with the computing platform, our main results must pass a replication rule.

II. Related Work

We review studies from 2022–2026 (Table I; a broader review is [5]). Second-life studies estimate the *current* SOH of a fixed set of batteries, from partial charging [1], health indicators [2], impedance spectra [3], statistical features with bagged GPR [4] or online clustering [6], with reported errors up to about 2.3% on their own data. None updates its model as new groups arrive.

Continual learning has been used to predict capacity loss [7] and, with progressive networks and TCNs, the state of charge across drive cycles, with attention to forgetting and task order [8]. Closest to our setting, pulse-test conditions have been learned one after another, with a replay buffer, to estimate second-life SOH with almost no forgetting [9]. Domain generalisation [10] and probabilistic BiLSTM/TCN models with isotonic recalibration [11] also target second-life SOH, but without a sequence of tasks.

TABLE I
Approaches of the reviewed studies and of this work. Their errors are not comparable with ours (other cells, targets and SOH definitions).

Study	Approach	Predicts	CL	Reuse
[1]	partial-charge features	current SOH, second life	–	–
[2]	58 health indicators	current SOH, second life	–	–
[3]	impedance (EIS) + ML	current SOH, second life	–	–
[4]	statistical features, bagged GPR	current SOH, screening	–	✓
[6]	online ML, clustering	current health, second life	–	–
[7]	bilevel-optimised CL	capacity degradation	✓	–
[8]	PNN + TCN	state of charge	✓	–
[9]	pulse features, replay CL	current SOH, second life	✓	–
[10]	domain generalisation	current SOH, second life	–	–
[11]	BiLSTM, TCN, TCN-GPR	SOH with uncertainty	–	✓
This work	residual TFT + EWC, shrinkage, conformal intervals	SOH 1–6 checks ahead, interval	✓	✓

Research gap. So far, continual learning for second-life SOH has *estimated* the current SOH [9]. To our knowledge, it has not been tested for *forecasting* across different groups of second-life batteries, together with reuse screening, a measure of forgetting, and a comparison with simple forecasters and with training on all data at once. We ask whether a continual learner can learn such groups one after another without losing the earlier ones, and whether its forecasts support reuse screening.

III. The Residual TFT-EWC Approach

We forecast the SOH one to three checks ahead (up to six in later experiments) from everything measured up to a reference performance test (RPT), i.e. a capacity check, with cells or usage conditions arriving as tasks. Estimating the *current* SOH failed on the first dataset: cycle-level features carried almost no health information (median within-cell Spearman correlation with SOH 0.09, against -0.99 for elapsed time), and gradient boosting and ridge (1.78, 2.37 pp) lost to persistence, which repeats the last measured SOH (0.99 pp). We therefore switched

TABLE II

DATASETS. S1 AND S6 FORM THE CONTINUAL-LEARNING STREAMS; S7–S9 FORM THE FIRST-LIFE PRETRAINING POOL; S7 AND S10 GIVE THE MANY-CONDITION TESTS.

Set	Cells and conditions	Label	Samples
S1 [12], [13]	6 NMC 21700 cells, second life on residential or commercial peak shaving, 20/35 °C blocks	C/20 RPT capacity, pulses excluded	279
S6 [14], [15]	20 LFP 18650 cells, 7 conditions (charge mode, voltage and rate, discharge, depth), 50 °C	C/25 check capacity	2,409
S7 [16], [17]	195 of 228 NMC/C+SiO cells, calendar, cyclic and WLTP ageing, 0–40 °C	check-up capacity	10,251
S8 [18], [19]	55 NCM cells, six charge/discharge protocols	capacity cycles	9,699
S9 [20], [21]	119 of 130 NCA, NCM and blended cells, 25–45 °C	capacity cycles	16,761
S10 [22], [23]	211 of 279 NMC 21700 cells, 71 calendar and cyclic conditions, 10–45 °C	check-up capacity	6,495

to forecasting after seeing the data, fixing the protocol before fitting any forecaster.

A. Datasets

Table II lists the six public datasets (all CC BY 4.0). Every file is checked against its repository record or checksums, and our parsing of the raw files exactly reproduces the processed S1 RPT capacities and the authors' capacities of all 843 S6 checks.

S1 contains six LG Chem INR21700-M50T cells that spent 23 months under an EV driving profile [24] and then about 27 months on peak-shaving duty cycles [25]; each RPT is a C/20 discharge interrupted by C/2 pulses, with impedance spectroscopy (EIS) from the fifth RPT [12]. RPT 2, run at about 20 °C, is dropped.

S6 was chosen by criteria fixed before the search (licence, a series of capacity checks, ageing past the end of first life, and the variety that S1 lacks). Its cells were checked every 100 cycles and end at 32–64% SOH; here "second life" means further ageing below 80% SOH under the same test. A check is used only if its C/25 discharge starts from full charge and reaches the cut-off voltage (843 of 847; Fig. 2). From the first-life sets *S7–S9* we keep cells with at least five valid checks (369 cells, 36,711 samples). *S10* uses the discharge capacity of each 23 °C check-up (2,587 of 2,766 kept).

B. Big-data processing of the ageing cycles

Between checks, every S6 cell logged each ageing cycle about once per second (827 files, 132 GB, 9.0×10^8 rows). Apache Spark 4.2 (PySpark, local mode) condenses them into the mean, last value and slope of seven measured quantities per 100-cycle block (21 inputs), each linked to the check that ends the block (details and validation in the supplement).

C. State of health and forecasting target

For cell b with check capacities $C_{b,1}, C_{b,2}, \ldots$ in time order,

$$\mathrm{SOH}_{b,k} = 100\, C_{b,k}/C_{b,1}\ \%, \tag{1}$$

the capacity-ratio definition [5], [26], here relative to the first valid measurement, not to the rated capacity. In S1 the capacity is the charge of the C/20 steps only [12, Eq. (1)]; the processed tables also counted the pulse charge, shifting some labels by up to 0.09 pp.

A sample is a triple (cell b, origin k, horizon $h \in \{1, 2, 3\}$): a forecast made at check k for check $k+h$, about one to three months ahead in S1 and 100–300 cycles in S6. The target is

$$\Delta_{b,k,h} = \mathrm{SOH}_{b,k+h} - \mathrm{SOH}_{b,k}, \tag{2}$$

the multi-horizon target of [27], so an error in Δ is the same as the error in the future SOH. Errors are root-mean-square errors (RMSE) in SOH percentage points (pp). The fade rate $r_{b,k}$ is the ordinary least-squares slope [28] of SOH against time t (days or cycles) over the last three valid checks,

$$r_{b,k} = \frac{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})(\mathrm{SOH}_{b,i} - \overline{\mathrm{SOH}})}{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})^2}. \tag{3}$$

Inputs use only what is known at the origin. **F0** contains the SOH history ($\mathrm{SOH}_{b,k}$, time or cycles since the first check, $r_{b,k}$), fixed descriptors (S1: duty cycle and first-life history [12]; S6: test conditions) and the *planned* usage until the target (S1: days at each temperature set point and at rest; S6: cycles), so forecasts assume the planned usage is known. **F1** adds diagnostics measured at the origin: 47 in S1 (pulse resistances, incremental-capacity and EIS features) and 17 in S6 (plateau charges, incremental-capacity peaks, hysteresis, a resistance proxy). Test currents are excluded: in S1 they were set from the previous capacity and would give away the label. Missing values are filled from training data and flagged.

D. Splits and protocols

Purged chronological splits. Each cell's checks are cut at 60% (A) and 80% (B). Training targets lie at or before A; validation samples start at or after A and end by B; test samples start at or after B. Samples that straddle a cut are dropped, so no training label falls in the validation period and no validation label in the test period.

P1, continual stream. A task is one cell in S1 (six tasks) or one test condition in S6 (seven tasks). At step i of an ordering π, a method sees only the data of task π_i. Every method runs on the same 10 task orders with 5 seeds.

P2, unseen group. One cell (S1) or condition (S6) is held out for testing, and models are trained on the rest.

P3, first-life pretraining. All datasets are mapped onto shared inputs (SOH history, calendar and cycle time, planned time to the target, operating conditions and chemistry; undefined conditions flagged as missing). Models are first trained on the pool, which also sets the input scaling and the first EWC anchor; the seven S6 conditions then follow as a stream.

E. Models

Preprocessing (standardisation, median imputation with missing indicators, target scaling) is fitted on the first task and then frozen. The *MLP* has hidden layers of 128 and 64 ReLU units (dropout 0.1). The proposed model builds

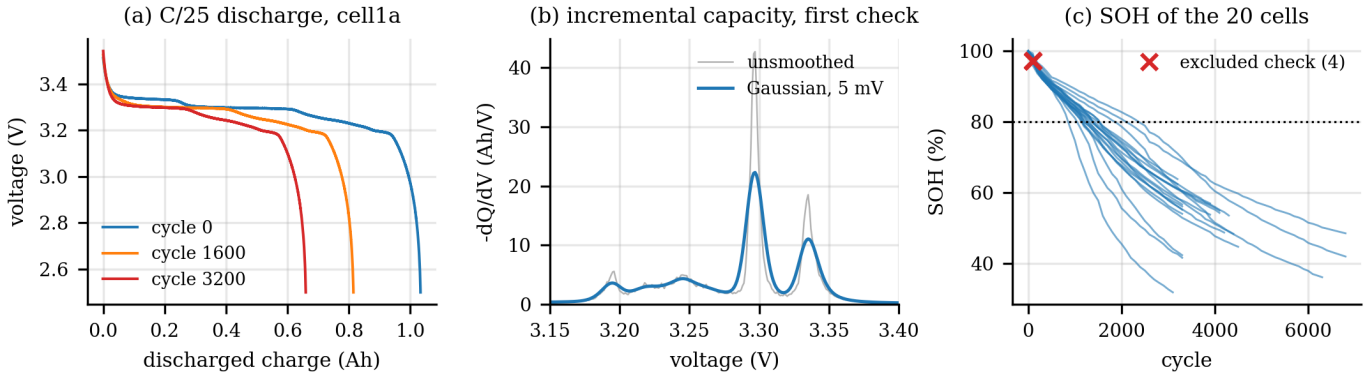


Fig. 2. Preprocessing of S6. (a) C/25 discharge of one cell at three ages. (b) Incremental capacity at the first check, before and after Gaussian smoothing (5 mV), the source of the peak inputs. (c) SOH (1) of the 20 cells over the 843 valid checks (crosses: the four excluded; dotted: 80%).

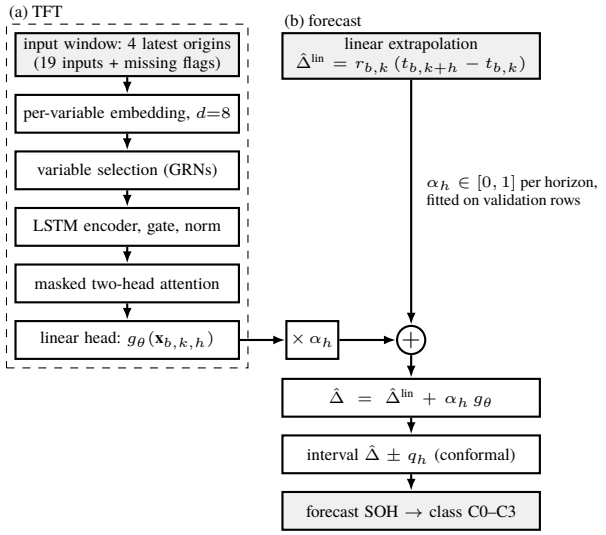


Fig. 3. Proposed model, Residual TFT-EWC. (a) The TFT maps a window of the four latest origins of a cell to a correction g_θ. (b) The correction is shrunk per horizon and added to linear extrapolation (5); conformal intervals give the reuse class. Each task is trained on the correction with the EWC penalty (4) (Fig. 1b).

on the Temporal Fusion Transformer (TFT) [27]: it looks at the $W{=}4$ most recent origins of the same cell and horizon (inputs only, never labels), which pass through per-variable embeddings, variable selection by gated residual networks, an LSTM encoder [29] and two-head attention restricted to past steps (Fig. 3; $d{=}8$, 12,708 parameters with the shared inputs).

Strategies. Fine-tuning trains on the current task only (a lower bound); experience replay [30] keeps B samples of each finished task; joint training fits all tasks at once (an upper bound); and Elastic Weight Consolidation (EWC) [31] penalises changes to the weights that mattered for each finished task j,

$$\frac{\lambda}{2}\sum_{p} F_{j,p}\left(\theta_p - \theta^*_{j,p}\right)^2, \tag{4}$$

the penalty of [31, Eq. (3)], where θ^*_j are the parameters after task j and F_j is the empirical diagonal Fisher information over up to 500 training samples. *TFT+EWC* is the proposed continual learner; trained on the residual (5) below, with shrinkage and intervals, it becomes the proposed *Residual TFT-EWC* (Fig. 3). Training uses Adam [32], batches of 64, at most 200 epochs and early stopping on the current task's validation RMSE.

Reference forecasters are persistence ($\hat{\Delta} = 0$, the naïve forecast of [33]), linear extrapolation of the least-squares fade rate (3) [28], $\hat{\Delta}_{b,k,h} = r_{b,k}(t_{b,k+h} - t_{b,k})$, ridge regression [28] (on planned usage only, F0 or F1) and histogram gradient boosting [34] on F1.

Learned residual. Because linear extrapolation proved hardest to beat, a follow-up experiment, registered in advance, trains the continual learners to correct it rather than replace it. With $\hat{\Delta}^{\text{lin}}_{b,k,h}$ the extrapolation above (zero when $r_{b,k}$ is missing), the model g_θ learns $\Delta_{b,k,h} - \hat{\Delta}^{\text{lin}}_{b,k,h}$ and the forecast is

$$\hat{\Delta}_{b,k,h} = \hat{\Delta}^{\text{lin}}_{b,k,h} + g_\theta(\mathbf{x}_{b,k,h}), \tag{5}$$

a residual hybrid of a simple and a learned forecaster, as in [35].

Shrinkage and intervals. After training, without changing the weights, the correction is scaled back per horizon by $\alpha_h = \min\big(1, \max(0, \sum p\,y / \sum p^2)\big)$, the least-squares slope through the origin [28] of the observed (y) on the predicted (p) correction on calibration rows. A split-conformal interval [36] $\hat{\Delta} \pm q_h$ takes as q_h the $\lceil (n+1)(1-a) \rceil$-th smallest of the n absolute calibration errors at horizon h. Intervals at level $1-a$ (90% and 80%) are scored by coverage, width and the interval score (Table IV; pp, lower is better). On the stream both are refitted after each task on its validation rows; extrapolation gets intervals the same way.

Registered experiments. Each follow-up experiment was registered before it was run (Table III). Origins with a fade rate (H^+) and without one (H^-, a cell's first two checks, where extrapolation equals persistence) are reported separately. Horizons are 1–3 (E1) and 1–6 (E2); a question counts only if it replicates in both.

Bounded-memory variant (V2). A last experiment, registered after the S7 and S10 test results had been read per condition (a

TABLE III
REGISTERED FOLLOW-UP EXPERIMENTS (ALL AFTER P1–P3).

Experiment	Setting	Question
Residual	S6 stream, P3 inputs, from scratch	residual vs. direct target; beats extrapolation?
SA-EWC	S6 stream	EWC penalty scaled by the input shift vs. fixed λ; BiLSTM [11], TCN [8], [11], Transformer, GPR [3], [4], random forest, SVR (supplement)
Shrinkage, intervals	S6 stream	shrunk residual model vs. extrapolation; interval coverage
Unseen conditions	S6, P2; six training conditions streamed in the ten orders; shrinkage and intervals fitted once on pooled validation rows	beats extrapolation at H^+; vs. gradient boosting and joint TFT
GB intervals	S6, P2; each training condition predicted by a model fitted on the other five (cross-validation interval [37], conditions as folds)	coverage on held-out conditions (no finite-sample guarantee)
Many conditions	S7 (67 conditions), S10 (71); seven tune, the rest (60, 64) form five test folds	beats GB refitted on the stream's last condition, GB on all conditions, extrapolation (Wilcoxon over held-out conditions); S7 shrinks at every origin, S10 only at H^+ (rule fixed after S7)

development test, not a confirmatory one), changes the residual TFT in three ways: (i) experience replay replaces EWC: 32 training and 8 validation rows of each finished condition are kept (second arm: 8 and 2), each mini-batch loss adds the squared error on an equally large batch of stored rows, and early stopping also uses the stored validation rows; (ii) the forecast is the mean of the three models of a seed set; (iii) α_h and the half-widths are fitted, per horizon and for H^+ and H^- separately, on the forecasts the ensemble made for each condition of the second half of the stream before training on it. Folds, orders and horizon sets are those of the many-condition tests, with seeds 0–2 (main) and 5–7 (replicate); only the learning rate is tuned. The registered questions are whether V2 is more accurate than gradient boosting with all conditions stored (Q1), than gradient boosting on the newest condition plus V2's stored rows (GB-ER32, Q2), than extrapolation at H^+ (Q3) and than the earlier model of Table VII (Q4) (Wilcoxon over held-out conditions, Holm correction, replication rule; a win also needs the median difference and most conditions to favour V2 in both seed sets), and whether its pooled 90% coverage reaches 0.85 (Q5); each must hold for horizons 1–3 and 1–6.

F. Hyperparameters, metrics and statistics

Tuning. Hyperparameters are tuned on validation data from orders never used for evaluation, in two steps: first the learning rate of each backbone ($3 \cdot 10^{-4}$–10^{-1}), then the replay buffer B (50–400) and the EWC strength λ (10^0–10^5). A grid is extended if its best value lies on an edge. From P3 on, ten tuning orders are used, and the smallest value within one standard error of the best is kept.

Metrics. After training step i, every task j is tested, giving the error matrix $R_{i,j} = \mathrm{RMSE}(\mathcal{D}^{\mathrm{te}}_{\pi_j} \mid \theta_i)$ [38] over T tasks. Table IV lists the metrics and their sources; positive BWT and forgetting mean forgetting. P2 adds standard error and class metrics, e.g. MAE, R^2, R^2_Δ of the change (≤ 0 for persistence),

TABLE IV
EVALUATION METRICS (BWT: BACKWARD TRANSFER)

Metric	Definition	Source
Average error	$\frac{1}{T}\sum_j R_{T,j}$	[38]
BWT	$\frac{1}{T-1}\sum_{j<T}(R_{T,j} - R_{j,j})$	[38]
Forgetting	$\frac{1}{T-1}\sum_{j<T}\left(R_{T,j} - \min_{j\leq i<T} R_{i,j}\right)$	[39]
Skill	$1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persist.}}$	[33]
Coverage	share of Δ inside $\hat{\Delta} \pm q_h$	[36]
Interval score	$2q_h + \frac{2}{a}\max(\lvert\Delta - \hat{\Delta}\rvert - q_h, 0)$	[40]
Reuse class	C0 >95, C1 90–95, C2 80–90, C3 $\leq$80 %	ours

MCC, Cohen's κ_w and the AUC of SOH$\leq$80% (supplement Tables SI–SII).

Statistics. Means carry 95% bootstrap intervals [41]. Methods are compared by two-sided Wilcoxon signed-rank tests [42] on matched runs or held-out groups, with rank-biserial effect sizes [43] and Holm correction [44]. A difference d counts as *replicated* only if, in both the main seed set (0–4) and a replicate set (5–9), it is significant, has the same sign and $|d| \geq 0.05$ pp. If $|d| < 0.05$ pp in both, the methods are practically equivalent. This threshold is about half the label shift caused by the pulse-charge definition.

G. Implementation and availability

The pipeline uses Python 3.11, NumPy 2.4, pandas 3.0, scikit-learn 1.9 and PyTorch 2.14.1 (P1 and P2 were first run on an Intel Xeon with an unrecorded PyTorch version and re-run on an Intel Core 5 210H), with one CPU thread and seeded, deterministic algorithms. Every result file records configuration, seeds, library versions and the Git commit, and unit tests cover input causality, split purging and label exclusion. The datasets are at doi:10.17605/OSF.IO/8JNR5 (S1), 10.57745/OLBXKT (S6), 10.35097/1969 (S7), 10.5281/zenodo.10963339 (S8), 10.5281/zenodo.6405084 (S9) and 10.6084/m9.figshare.25975315 (S10). Code, decision log, preregistrations and results: https://github.com/TarunSitaraman/continual-learning-heterogeneous-second-life-ev-batteries.

IV. RESULTS

Summary. The proposed model is the only one that does well in both settings (Table V, Fig. 4a,b; pairs such as 0.32/0.31 give the main and replicate seed sets). On the stream it has the lowest error for horizons 1–3, level with extrapolation, and is 0.13–0.41 pp more accurate than gradient boosting (replicated; not planned in advance for the shrunk model). On an unseen condition it is 0.43–0.68 pp more accurate than extrapolation (replicated), and gradient boosting with every condition stored is best. Held to our rule (one condition at a time, no earlier data kept), gradient boosting is slightly better on unseen conditions for horizons 1–3 (0.06/0.08 pp, replicated) and level for 1–6, but on the stream it forgets (0.17–0.34 pp) and is 0.24–0.46 pp worse (replicated, post hoc). Only the proposed model is within 0.07 pp of the best in both settings.

TABLE V

THE PROPOSED MODEL AGAINST THE TWO STRONGEST TRADITIONAL FORECASTERS ON S6, IN BOTH SETTINGS. TOP: TEST RMSE (PP); PROPOSED MODEL: MEAN OVER 50 RUNS, MAIN/REPLICATE SEED SETS; OTHERS: SINGLE DETERMINISTIC FITS (BOLD: BEST). BOTTOM: PROPOSED MINUS BASELINE (PP, MAIN/REPLICATE; NEGATIVE FAVOURS THE PROPOSED MODEL) AND THE VERDICT OF THE REPLICATION RULE (BETTER/WORSE: SIGNIFICANT AFTER HOLM IN BOTH SEED SETS, SAME SIGN, AT LEAST 0.05 PP; EQUIV.: BELOW 0.05 PP IN BOTH; N.R.: NOT REPLICATED). *POST HOC FOR THE SHRUNK MODEL; REGISTERED FOR THE UNSHRUNK MODEL (D23), WITH THE SAME VERDICT. †UNDER THE CONTINUAL CONSTRAINT OF THE PROPOSED MODEL (CONDITIONS ONE AT A TIME, EARLIER DATA NOT KEPT): REFITTED ON THE NEWEST CONDITION (P2: D32; P1: D35, RESIDUAL TARGET, THE MORE ACCURATE OF ITS TWO FORMS). ‡POST HOC (D35).

	Continual stream (P1)		Unseen usage condition (P2)	
	h=1–3	h=1–6	h=1–3	h=1–6
Residual TFT-EWC (proposed)	**0.32/0.31**	0.50/0.49	0.92/0.94	1.62/1.65
Linear extrapolation	0.34	**0.48**	1.37	2.30
Gradient boosting, all conditions stored	0.45	0.90	**0.54**	**1.05**
Gradient boosting, latest condition only†	0.56	0.96	0.86	1.59
Persistence	1.64	2.67	2.82	4.90
Proposed minus baseline, verdict				
vs. linear extrapolation	−0.02/−0.04 equiv.	+0.01/+0.01 equiv.	−0.45/−0.43 better	−0.68/−0.65 better
vs. gradient boosting, all stored	−0.13/−0.14 better*	−0.40/−0.41 better*	+0.39/+0.41 worse	+0.57/+0.60 worse
vs. gradient boosting, latest only†	−0.24/−0.25 better‡	−0.46/−0.46 better‡	+0.06/+0.08 worse	+0.03/+0.06 n.r.

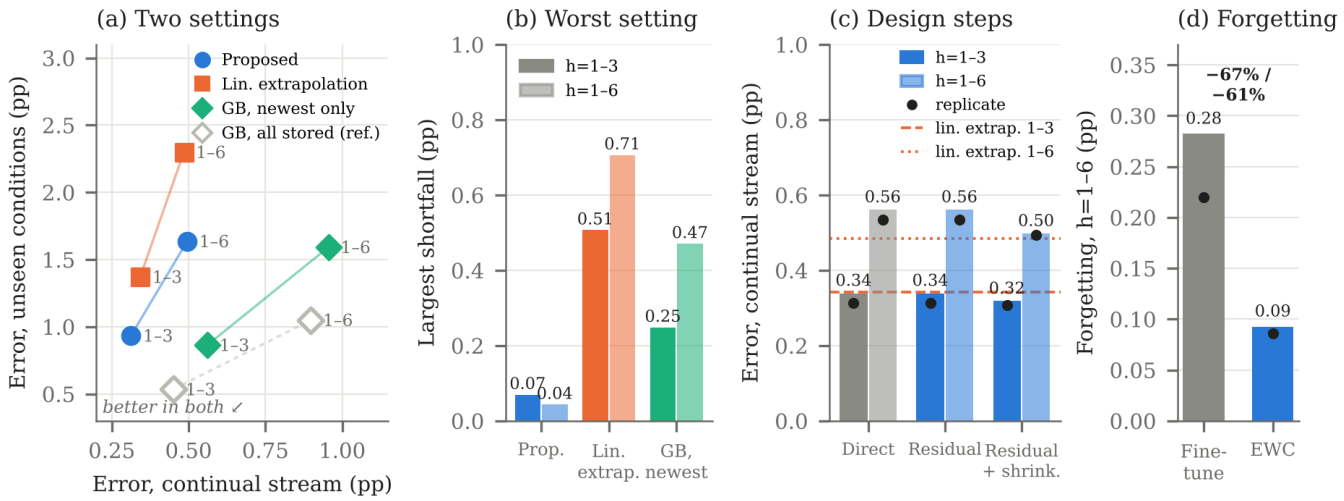


Fig. 4. S6. (a) Stream (P1) against unseen-condition (P2) error, horizons 1–3 and 1–6, for methods that keep no earlier data; hollow grey: gradient boosting with all conditions stored; persistence (1.6–4.9 pp) off scale. (b) Largest shortfall against the best of these in either setting. (c) Stream error by design step (dots: replicate seeds; lines: extrapolation). (d) Forgetting, horizons 1–6. GB on the stream: post hoc. Per-backbone errors: Fig. S10, Table SIII.

Data. The S1 diagnostics follow the current SOH closely (median within-cell $\rho = 0.97$ for the main incremental-capacity peak) but say little about the change to the next RPT (median $|\rho| \leq 0.3$). Every S1 sample stays in class C0, while all S6 cells pass from C0 to C3.

A. Continual stream (P1)

S1. All methods perform about the same: TFT + EWC reaches 0.462 pp, level with gradient boosting, fine-tuning and MLP replay (0.466–0.475 pp; Table SIV of the supplement), and forgetting is small (0.016–0.040 pp).

S6. When tasks differ in usage, forgetting is large (0.18–0.46 pp). EWC lowers the TFT's error against fine-tuning (0.677 against 0.883 pp; significance changed between runs, §IV-B), the MLP beats the TFT (0.531–0.540 pp), and extrapolation (0.343 pp) and gradient boosting (0.372 pp) are best.

B. Unseen cell or usage condition (P2)

Gradient boosting is best on both datasets (Table SVII of the supplement). On S6 the 100% depth-of-discharge condition is the hardest for every method except extrapolation. With six or seven held-out groups, no difference from the TFT survives Holm correction.

Gradient boosting also has the lowest MAE and the largest R^2_Δ (supplement), and the S6 reuse classes give the same ranking (Table VI).

Reproducibility. A re-run of P1 and P2 in a fixed software environment, on data rebuilt from the raw archives, kept the baselines and ranking but moved the neural models by up to 0.059 pp, changing the significance of differences of 0.005–0.014 pp; hence the stricter tuning and replication rule from P3 on.

TABLE VI

S6, P2: REUSE CLASS (C0–C3) AND END OF FIRST LIFE (EoL, SOH $\leq$ 80) OF THE FORECAST SOH, HELD-OUT SAMPLES POOLED, MEAN OVER SEEDS; F1 MACRO, κ_w LINEAR-WEIGHTED. ALL METRICS AND CLASS COUNTS: TABLE SII OF THE SUPPLEMENT.

Method	Acc.	F1	MCC	κ_w	EoL F1	EoL AUC
Persistence	0.858	0.661	0.723	0.809	0.962	0.998
Linear extrapolation	0.939	0.777	0.876	0.909	0.989	0.999
Ridge, usage only	0.954	0.851	0.905	0.932	0.989	0.999
Ridge, F0	0.963	0.881	0.922	0.942	0.990	0.999
Ridge, F1	0.958	0.852	0.912	0.935	0.986	0.999
Gradient boosting, F1	0.966	0.893	0.928	0.948	0.989	0.999
MLP	0.962	0.871	0.921	0.942	0.989	0.999
TFT	0.963	0.882	0.923	0.943	0.989	0.999

C. First-life pretraining (P3)

Pretraining on the first-life pool (36,711 samples, against 2,409 in S6) does not help: the TFT and the MLP are better trained from scratch, and linear extrapolation stays the most accurate method (supplement).

Why the TFT improved. A later ablation, repeated on the replicate seeds, explains the TFT's gain from 0.891 to 0.401 pp: dropping the 17 diagnostic features (0.17–0.35 pp) and adding the calendar time between checks (0.19–0.31 pp), both replicated.

D. Learned residual on linear extrapolation

Learning the residual (5) (Fig. 4c) did *not* meet its registered criterion of beating extrapolation: the two are practically equivalent for horizons 1–3, and for 1–6 extrapolation is better at h = 1–2 (replicated). It does make the TFT competitive: 0.17–0.30 pp more accurate than the direct form and ahead of gradient boosting (replicated). Over horizons 1–6, EWC cuts its forgetting from 0.28/0.22 to 0.09 pp (Fig. 4d, replicated; its effect on the final error does not replicate). The 21 Spark inputs make it worse (0.480/0.501 against 0.339/0.314 pp, replicated). *SA-EWC and literature backbones.* SA-EWC and a fixed λ are practically equivalent, and no literature backbone beats the residual TFT for both horizon sets (supplement).

E. Shrinkage and intervals on the stream

Shrinking the correction makes the model practically equivalent to extrapolation at every horizon (0.318/0.308 against 0.343 pp for horizons 1–3, 0.498/0.493 against 0.485 pp for 1–6), short of the registered criterion; its 90% intervals cover too little (0.71–0.81). Pretraining on the first-life pool makes it worse (0.22 pp, replicated).

F. Residual model on unseen conditions

On unseen conditions (Table V; Table SV of the supplement), the shrunk model beats extrapolation at H^+ origins, as registered (0.12–0.20 pp, replicated); at H^- origins extrapolation reduces to persistence and loses by over 2 pp. Gradient boosting (0.39–0.60 pp, on six of the seven conditions) and the joint TFT (0.32–0.44 pp) are more accurate (replicated). The model puts 95.5% of held-out samples in their measured reuse class (extrapolation 93.9%, gradient boosting 97.7%), every error off by one class.

TABLE VII

UNSEEN USAGE CONDITIONS UNDER THE CONTINUAL CONSTRAINT: S7 (60 TEST CONDITIONS) AND S10 (64). TOP: MEAN TEST RMSE OVER CONDITIONS (PP); PROPOSED: MAIN/REPLICATE SEEDS, SHRUNK ON S7 AT EVERY ORIGIN, ON S10 AT H^+ ONLY; JOINT TFT: MAIN SET; GB: GRADIENT BOOSTING. BOTTOM: PROPOSED MINUS BASELINE (PP, MAIN SET, REPLICATE WITHIN 0.04 PP) AND VERDICT (AS TABLE V). ‡MEAN SET BY A FEW CALENDAR CONDITIONS; GB IS BETTER ON 52 OF 64.

	S7 (D34)		S10 (D36)	
	h=1–3	h=1–6	h=1–3	h=1–6
Proposed	5.20/5.18	7.49/7.48	1.04/1.04	1.54/1.55
Lin. extrapolation	5.31	7.90	1.21	1.90
GB, latest only	6.50	8.40	1.88	2.38
GB, all stored	3.85	5.12	1.39	4.77
Joint residual TFT	4.12	5.79	0.81	0.96
Persistence	6.61	9.11	1.39	2.18
vs. GB, latest only	−1.30 better	−0.91 n.r.	−0.84 better	−0.84 better
vs. lin., H^+	−0.01 equiv.	−0.07 n.r.	−0.07 better	−0.19 n.r.
vs. GB, all stored	+1.36 worse	+2.37 worse	−0.35 better‡	−3.23 n.r.

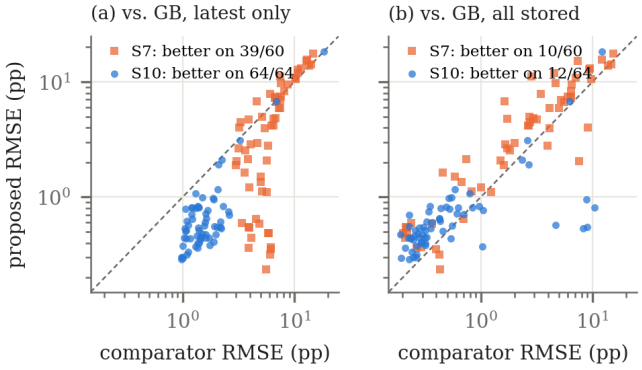


Fig. 5. Per-condition error on S7 and S10 (horizons 1–3, main seeds). Points below the diagonal favour the proposed model.

Intervals. Calibrated on the training conditions, 90% intervals cover too little on the held-out one (0.66–0.68 for the TFT, 0.74–0.76 for extrapolation). Calibrated by leaving one condition out, gradient boosting reaches the registered 0.85 (0.90), except on the 100% depth-of-discharge condition, outside the others' range (0.48, 0.38). Refitting it at each new check lowers its interval score by 29%.

G. Many unseen conditions (S7, S10)

Under the continual constraint, the proposed model beats gradient boosting refitted on the latest condition (Table VII, Fig. 5): on S7 for horizons 1–3 only (1.30–1.32 pp; for 1–6 not replicated, $p_{\text{Holm}} = 0.10$), and on S10, shrinking only at H^+ origins, on all 64 conditions for both horizon sets (0.83–0.85 pp, $p_{\text{Holm}} < 10^{-10}$), its first registered win over a traditional forecaster. Gradient boosting with every condition stored remains better on most conditions (Fig. 5b); its higher S10 mean comes from a few calendar conditions where it fails (up to 17 pp), so the S10 verdict for horizons 1–3 is not counted as a win.

TABLE VIII

V2 (replay of 32 rows per condition, three-seed ensemble, predict-before-training calibration) on the unseen conditions of Table VII; post hoc development test. Top: mean test RMSE over conditions (pp), main/replicate seeds (GB: one deterministic fit); earlier model: shrunk at H^+ only (S10), at every origin (S7); joint TFT: committed D34/D36 values. Bottom: V2 minus comparator (pp, main set, replicate within 0.08 pp), verdict, and Q5. Every verdict is replicated.

	S10		S7	
	h=1–3	h=1–6	h=1–3	h=1–6
V2	0.78/0.80	1.01/1.00	4.12/4.11	5.89/5.90
GB, all stored	1.39	4.77	3.85	5.12
GB-ER32	1.55	5.19	3.73	5.03
Lin. extrapolation	1.21	1.90	5.31	7.90
Earlier model	1.04/1.04	1.54/1.55	5.20/5.18	7.49/7.48
Joint residual TFT	0.81/0.81	0.96/1.01	4.12/4.33	5.79/6.18
Q1 vs. GB, all stored	-0.61 better	-3.76 better	$+0.28$ worse	$+0.77$ worse
Q2 vs. GB-ER32	-0.77 better	-4.17 better	$+0.39$ worse	$+0.86$ worse
Q3 vs. lin., H^+	-0.22 better	-0.56 better	-0.20 better	-0.45 better
Q4 vs. earlier model	-0.26 better	-0.53 better	-1.08 better	-1.60 better
Q5 90% coverage	0.91/0.90	0.89/0.90	0.90/0.91	0.89/0.90

H. Bounded memory: the replay ensemble V2 (S7, S10)

On S10, V2 answers all five registered questions positively (Table VIII); on S7, Q3–Q5 are positive and Q1 and Q2 negative, as gradient boosting with all conditions stored and GB-ER32 are better (replicated). The S10 gains for Q1 and Q2 come mostly from the conditions where gradient boosting fails (all stored, horizons 1–6: mean 4.77, median 0.83 pp): the median differences are small (Q1: -0.02 to -0.03 pp for horizons 1–3, -0.13 to -0.14 pp for 1–6; Q2: -0.05 to -0.06 and -0.18 to -0.19 pp), and gradient boosting with all conditions stored is better on 22–23 of 64 conditions for 1–3. On S7 it is better on 43–51 of 60 conditions, most at H^- origins, and the two boosting comparators are better by 0.27–0.87 pp on average. V2 shows no replicated difference from the joint TFT, which stores all data. Replay gives most of the gain over the EWC model (0.26–0.81 pp, replicated); the ensemble adds 0.05–0.14 pp, and calibration leaves the error unchanged. Coverage is uneven by ageing type: 0.82/0.83 on S10 calendar conditions for horizons 1–6, 0.81–0.84 on S7 cyclic ones.

V. Discussion

Where the proposed model is strongest. Extrapolation trails on new conditions and gradient boosting on the stream; Residual TFT-EWC holds up in both. Matching extrapolation is itself hard: fade is smooth and tasks are small (21–27 and 135–225 training samples in S1 and S6). Most of the gap to gradient boosting and joint training on unseen conditions comes from storing earlier data, not from the model (Section IV-G): storing 40 rows per condition leaves the replay variant V2 with no replicated difference from joint training on S7 and S10, though gradient boosting stays better on S7 (Section IV-H). Where earlier data cannot be kept (privacy, ownership, volume), continual learning is the natural choice; here, storing every group was still cheap.

What drives its accuracy. Building on the best simple forecaster mattered most. The ablation points to the inputs, not the architecture: the calendar time between checks helps because at 50 °C all 20 S6 cells lose more capacity per cycle when checks are further apart.

Reuse. The reuse classes screen on relative capacity alone, while reuse also depends on power, safety, absolute capacity and cost. For a group unlike any seen before, such as the 100% depth-of-discharge condition, neither forecasts nor intervals can be trusted, and pooled coverage hides this, as it hides V2's shortfall on S10 calendar and S7 cyclic conditions.

Limitations. We switched to forecasting after inspecting the first dataset, and decided the ablation and all later experiments after the main results (all but the ablation registered before being run; the stream test of gradient boosting refitted on the newest condition is post hoc; S7 and S10 were added to find a setting where the model wins, the S10 rule after the S7 results; V2 was designed after their test results had been read per condition, so it is a development test, and a confirmatory test needs data not used so far). The second-life datasets are small; S6 is aged at 50 °C and its "second life" continues the same test; SOH is relative to each cell's first valid measurement; forecasts assume the planned usage is known. The TFT settings for unseen conditions were tuned on validation rows of all seven S6 conditions, from one laboratory, and leave-one-condition-out intervals cover only gradient boosting and extrapolation.

VI. Conclusion

We proposed Residual TFT-EWC, a continual learner that forecasts second-life SOH as a scaled-down correction to linear extrapolation. It is the only model compared that holds up both on the stream (level with extrapolation, ahead of gradient boosting, far less forgetting with EWC) and on unseen conditions (ahead of extrapolation, and of gradient boosting held to the same rule on all 64 unseen S10 conditions). Gradient boosting and joint training, which keep all earlier data, stay more accurate there. A post hoc replay variant that keeps 40 rows per condition shows no replicated difference from the joint TFT on S7 and S10 and beats gradient boosting on S10 on average, mostly where boosting fails, but not on S7. Intervals held only when calibrated on conditions left out of training, and not outside the training range. Future work should test settings where data cannot be stored and groups are larger, and add power and safety criteria to the screening.

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